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1 Problem 1.23 DONE

1.1 Task

(Norm Equivalences in $\mathbb{R}^n$). Show that $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are equivalent norms in $\mathbb{R}^n$ (of course, without using Theorem 1.56!).

1.2 Solution

Two norms $\|\cdot\|_a, \|\cdot\|_b$ are equivalent if there exist $c, C > 0$ with $c\|x\|_a \leq \|x\|_b \leq C\|x\|_a$ for all $x \in \mathbb{R}^n$. Let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.

- $\|\cdot\|_\infty$ and $\|\cdot\|_1$:

  Each $|x_i| \leq \sum_j |x_j|$, so $\|x\|_\infty \leq \|x\|_1$. Conversely, each $|x_i| \leq \|x\|_\infty$, so $\|x\|_1 \leq n\|x\|_\infty$.

- $\|\cdot\|_\infty$ and $\|\cdot\|_2$:

  We have $\|x\|_\infty^2 = \max_i |x_i|^2 \leq \sum_i |x_i|^2 = \|x\|_2^2$, so $\|x\|_\infty \leq \|x\|_2$. Conversely, $\sum_i |x_i|^2 \leq n\|x\|_\infty^2$, so $\|x\|_2 \leq \sqrt{n}\,\|x\|_\infty$.

- $\|\cdot\|_1$ and $\|\cdot\|_2$:

  Since norm equivalence is transitive, $\|\cdot\|_1 \sim \|\cdot\|_\infty \sim \|\cdot\|_2$ implies $\|\cdot\|_1 \sim \|\cdot\|_2$. ■

2 Problem 1.38 DONE

2.1 Task

(Distances are not always induced by norms)

From Proposition 1.53 we have seen that for any normed vector space $(\mathbb{E}, \|\cdot\|_\mathbb{E})$, we can define a distance on $\mathbb{E}$ $d_{\|\cdot\|_\mathbb{E}}(x, y) = \|x - y\|_\mathbb{E}$ such that $(\mathbb{E}, d_{\|\cdot\|_\mathbb{E}})$ is a metric space. We shall see with this exercise that not every distance can be expressed in terms of a norm $\|x - y\|$.

(i) Let X be an arbitrary non-empty set, we define the *discrete distance* $\rho : X \times X \to \mathbb{R}$, given by

$$\rho(x, y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}.$$

Prove that (X, ρ) is a metric space.

(ii) We now set $X = \mathbb{R}$ with the distance $\rho : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ as defined above, prove by contradiction that ρ cannot be written as $\rho(x, y) = \|x - y\|$, for any $\|\cdot\| : \mathbb{R} \to \mathbb{R}$ defined

on $\mathbb{R}$. (**Indication:** Suppose that $\rho(x,y) = \|x-y\|$ for some $\|\cdot\|$, then $\|1\| = \rho(0,1) = 1$, and $\|2\| = |2| \cdot \|1\| = 2$. What does this contradict?)

(iii) Similarly to the previous point, prove this generalized result:

Given $\mathbb{E} \neq \{0\}$ a $\mathbb{R}$ vector space and $d(\cdot,\cdot) : \mathbb{E} \times \mathbb{E} \to \mathbb{R}$ a *bounded* metric (i.e. there exists $K > 0$ such that $d(x,y) \leq K$, $\forall x, y \in \mathbb{E}$). Then $d(\cdot,\cdot)$ cannot be written as $d(x,y) = \|x-y\|$, for any $\|\cdot\| : \mathbb{E} \to \mathbb{R}$.

(iv) Prove that for any norm $\|\cdot\| : \mathbb{E} \to \mathbb{R}$ defined on a $\mathbb{R}$ vector space $\mathbb{E} \neq \{0\}$, the function $d(\cdot,\cdot) : \mathbb{E} \times \mathbb{E} \to \mathbb{R}$ given by

$$d(x,y) := \frac{\|x-y\|}{1+\|x-y\|}$$

defines a distance on $\mathbb{E}$ but is not induced by a norm. (**Indication:** to see that $d(x,y)$ as defined above satisfies the triangular inequality, it might be useful to first prove that the function $\phi(t) = t/(1+t)$ is monotone increasing on the positive real numbers and subadditive, i.e. $\phi(a+b) \leq \phi(a) + \phi(b)$, for all $a, b \geq 0$)

2.2 Solution

2.2.1 (i)

We verify the three metric axioms for ρ.

- *Definiteness*: By definition, $\rho(x,y) = 0 \iff x = y$.
- *Symmetry*: If $x = y$ then $\rho(x,y) = 0 = \rho(y,x)$. If $x \neq y$ then $y \neq x$, so $\rho(x,y) = 1 = \rho(y,x)$.
- *Triangle inequality*: Let $x, y, z \in X$. If $x = z$, then $\rho(x,z) = 0 \leq \rho(x,y) + \rho(y,z)$. If $x \neq z$, then at least one of $x \neq y$ or $y \neq z$ holds (otherwise $x = y = z$), so $\rho(x,y) + \rho(y,z) \geq 1 = \rho(x,z)$.

Hence (X, ρ) is a metric space. ■

2.2.2 (ii)

Suppose for contradiction that $\rho(x,y) = \|x-y\|$ for some norm $\|\cdot\|$ on $\mathbb{R}$. Then $\|1\| = \rho(0,1) = 1$. By absolute homogeneity of the norm, $\|2\| = |2| \cdot \|1\| = 2$. But $\|2\| = \|2-0\| = \rho(2,0) = 1$, since $2 \neq 0$. This gives $2 = 1$, a contradiction. ■

2.2.3 (iii)

Suppose for contradiction that $d(x,y) = \|x-y\|$ for some norm $\|\cdot\|$ on $\mathbb{E}$. Since $\mathbb{E} \neq \{0\}$, there exists $v \in \mathbb{E}$ with $v \neq 0$, so $\|v\| > 0$. By absolute homogeneity, for any $n \in \mathbb{N}$:

$$d(nv, 0) = \|nv\| = n\|v\| \to \infty \quad \text{as } n \to \infty.$$

But d is bounded by K, so $n\|v\| \leq K$ for all n, which is impossible. ■

2.2.4 (iv)

Let $\phi(t) = t/(1+t)$ for $t \geq 0$, so that $d(x, y) = \phi(\|x - y\|)$.

Monotonicity: For $0 \leq a \leq b$, we have $\phi(b) - \phi(a) = (b-a)/((1+a)(1+b)) \geq 0$, so ϕ is increasing.

Subadditivity: For $a, b \geq 0$:

$$\phi(a) + \phi(b) - \phi(a+b) = \frac{a}{1+a} + \frac{b}{1+b} - \frac{a+b}{1+a+b}.$$

Bringing to a common denominator and expanding, the numerator becomes $a^2b + ab^2 + 2ab \geq 0$, so $\phi(a+b) \leq \phi(a) + \phi(b)$.

We now verify the metric axioms for d.

- *Identity of indiscernibles*: $d(x, y) = \phi(\|x - y\|) = 0 \iff \|x - y\| = 0 \iff x = y$, since $\phi(t) = 0 \iff t = 0$.
- *Symmetry*: $d(x, y) = \phi(\|x - y\|) = \phi(\|y - x\|) = d(y, x)$.
- *Triangle inequality*: Using the triangle inequality for $\|\cdot\|$, monotonicity, and subadditivity of ϕ:

$$d(x, z) = \phi(\|x - z\|) \leq \phi(\|x - y\| + \|y - z\|) \leq \phi(\|x - y\|) + \phi(\|y - z\|) = d(x, y) + d(y, z).$$

So d is a metric. Since $d(x, y) = \phi(\|x - y\|) < 1$ for all x, y, the metric d is bounded (with $K = 1$). By part (iii), d cannot be induced by any norm. ■

3 Problem 1.44 DONE

3.1 Task

> (Dual space). The following is a similar but perhaps more laborious exercise to the previous one (ex. 1.43).

Let $\mathbb{E} = \mathbb{C}_2[X]$ be the space of **complex** polynomials $p(X)$ with $\deg p \leq 2$ (suppose that $\{1, X, X^2\}$ is proven to be a basis on $\mathbb{E}$), and let $f, g \in \mathbb{E}^*$ be as defined below:

$$f(p(X)) = p(1), \qquad \text{and} \qquad g(p(X)) = p(i)$$

where $i \in \mathbb{C}$ is the imaginary unit element.

(i) Are f and g linearly independent?

(ii) If the previous point is affirmative, give a third element $h \in \mathbb{E}^*$ such that $\{f, g, h\}$ forms a basis on $\mathbb{E}^*$.

(iii) Does the system of equations

$$\begin{cases} f(p(X)) = i, & \text{and} \\ g(p(X)) = 1 \end{cases}$$

have any solution for $p \in \mathbb{E}$? if there is, give all the polynomials that are solution of the system. Is the solution unique?

3.2 Solution

Write $p(X) = a + bX + cX^2$ with $a, b, c \in \mathbb{C}$ throughout.

3.2.1 (i)

Suppose $\alpha f + \beta g = 0$ in $\mathbb{E}^*$, i.e. $\alpha p(1) + \beta p(i) = 0$ for all $p \in \mathbb{E}$. Evaluating on basis elements:

- $p = 1$: $\alpha + \beta = 0$,
- $p = X$: $\alpha + i\beta = 0$.

From the first equation $\alpha = -\beta$. Substituting into the second: $-\beta + i\beta = \beta(i - 1) = 0$, so $\beta = 0$ and thus $\alpha = 0$. Hence f and g are linearly independent. ■

3.2.2 (ii)

Since $\dim \mathbb{E} = 3$, we have $\dim \mathbb{E}^* = 3$, so we need one more functional. Define $h \in \mathbb{E}^*$ by $h(p(X)) = p(0)$ (evaluation at 0, i.e. $h(a + bX + cX^2) = a$).

We check that $\{f, g, h\}$ is linearly independent. Suppose $\alpha f + \beta g + \gamma h = 0$. Evaluating on basis elements:

- $p = 1$: $\alpha + \beta + \gamma = 0$,

- $p = X$: $\alpha + i\beta = 0$,
- $p = X^2$: $\alpha - \beta = 0$.

From the third equation $\alpha = \beta$. Substituting into the second: $\beta + i\beta = \beta(1+i) = 0$, so $\beta = 0$, hence $\alpha = 0$, and from the first $\gamma = 0$. Since three linearly independent vectors in a 3-dimensional space form a basis, $\{f, g, h\}$ is a basis of $\mathbb{E}^*$. ■

3.2.3 (iii)

The system $f(p) = i$, $g(p) = 1$ reads:

$$a + b + c = i, \qquad a + ib - c = 1.$$

Adding the two equations: $2a + (1+i)b = 1 + i$, so $a = \frac{(1+i)-(1+i)b}{2} = \frac{(1+i)(1-b)}{2}$. Subtracting the first from the second: $(i-1)b - 2c = 1 - i$, so $c = \frac{(i-1)b-(1-i)}{2} = \frac{(i-1)(b-1)}{2}$.

Let $b = t \in \mathbb{C}$ be a free parameter. Then:

$$a = \frac{(1+i)(1-t)}{2}, \qquad b = t, \qquad c = \frac{(i-1)(t-1)}{2},$$

and the general solution is:

$$p(X) = \frac{(1+i)(1-t)}{2} + tX + \frac{(i-1)(t-1)}{2}X^2, \qquad t \in \mathbb{C}.$$

The system has infinitely many solutions (a one-parameter family), so the solution is not unique. This is expected: two linear equations on a 3-dimensional space generically constrain a 1-dimensional affine subspace. ■

4 Problem 2.4 DONE

4.1 Task

(Inner product of functions)

We define the following set of functions:

$$\mathbb{E} = \{f : [0, \pi] \to \mathbb{R} \ : \ f(x) = \alpha \cos x + \beta \sin x, \ \alpha, \beta \in \mathbb{R}\}$$

(i) Prove that $\mathbb{E}$, when equipped with the usual addition ($(f+g)(x) = f(x) + g(x)$) and scalar multiplication of functions ($(\lambda \cdot f)(x) = \lambda \cdot f(x)$), is a vector space.

(ii) In the vector space $\mathbb{E}$ we define the following inner product:

$$\omega(f, g) = \int_0^\pi f(x)g(x)\,\mathrm{d}x$$

Is the basis $\mathcal{B} := \{\cos(x), \sin(x)\}$ an orthogonal basis w.r.t. ω? Is it orthonormal? Give an orthogonal/orthonormal basis if $\mathcal{B}$ is not.

4.2 Solution

4.2.1 (i)

It suffices to show that $\mathbb{E}$ is a subspace of the $\mathbb{R}$ vector space of all functions $[0, \pi] \to \mathbb{R}$ (which is a vector space under pointwise addition and scalar multiplication). We check the three subspace conditions.

- *Non-empty*: The zero function $0 = 0 \cdot \cos x + 0 \cdot \sin x$ belongs to $\mathbb{E}$.
- *Closure under addition*: If $f = \alpha_1 \cos x + \beta_1 \sin x$ and $g = \alpha_2 \cos x + \beta_2 \sin x$ are in $\mathbb{E}$, then $(f + g)(x) = (\alpha_1 + \alpha_2) \cos x + (\beta_1 + \beta_2) \sin x \in \mathbb{E}$.
- *Closure under scalar multiplication*: If $f = \alpha \cos x + \beta \sin x \in \mathbb{E}$ and $\lambda \in \mathbb{R}$, then $(\lambda f)(x) = (\lambda\alpha) \cos x + (\lambda\beta) \sin x \in \mathbb{E}$.

Hence $\mathbb{E}$ is a vector space. ■

4.2.2 (ii)

We compute the three relevant integrals.

$\omega(\cos, \cos) = \int_0^\pi \cos^2 x\, dx = \int_0^\pi \frac{1+\cos 2x}{2}\, dx = \frac{\pi}{2}$

$\omega(\sin, \sin) = \int_0^\pi \sin^2 x\, dx = \int_0^\pi \frac{1-\cos 2x}{2}\, dx = \frac{\pi}{2}$

$\omega(\cos, \sin) = \int_0^\pi \cos x \sin x\, dx = \int_0^\pi \frac{\sin 2x}{2}\, dx = \left[-\frac{\cos 2x}{4}\right]_0^\pi = \frac{-1+1}{4} = 0$

Since $\omega(\cos, \sin) = 0$, the basis $\mathcal{B} = \{\cos x, \sin x\}$ is orthogonal. It is not orthonormal, since $\omega(\cos, \cos) = \omega(\sin, \sin) = \pi/2 \neq 1$. Normalizing each basis vector, the orthonormal basis is:

$$\mathcal{B}' = \left\{ \sqrt{\frac{2}{\pi}} \cos x,\ \sqrt{\frac{2}{\pi}} \sin x \right\}. \quad ■$$